

Hammad Ahmad

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TECHNICAL SKILLS

Machine Learning, NLP & LLMs : PyTorch, scikit-learn, XGBoost, sentence-transformers, fastembed (ONNX), pgvector, semantic search, Retrieval-Augmented Generation (RAG), cross-encoder re-ranking, RAGAS, Anthropic Claude API, Ollama
Engineering & Data : Python, TypeScript, SQL, PL/pgSQL, FastAPI, asyncio, Flask, REST APIs, PostgreSQL full-text search, Supabase, Neo4j, Next.js, React, pandas, NumPy
Infrastructure & Quality : AWS (App Runner, Amplify, CloudFront), Oracle Cloud, Docker, Git, Linux, CI build gates, pytest, Playwright, Sentry

EXPERIENCE

Jobzyl | jobzyl.com

May 2026 - Present

Founder & Sole Engineer

Bradford, UK

- Build and operate a live, public job search aggregator end to end: 29 provider integrations and 6 applicant tracking systems, a 3.4M row Postgres index across 26 countries, FastAPI backend, Next.js frontend and AWS deploy pipeline.
- Shipped semantic CV-to-posting matching in production: 384-dimension multilingual sentence embeddings over pgvector against the full corpus, gated so a posting naming no recognised skill reports as too thin to score rather than a confident 0%.
- Built the LLM layer on Anthropic Claude (CV scoring, cover letter, interview prep), with prompt-injection defences on every call, per-user quotas and documented fail-open behaviour.
- Trained a pay regression model against the shipped salary benchmark as baseline, split by employer group to prevent leakage: MAE 27,420 to 24,001, MdAPE 18.9% to 16.4%. Reported per country and seniority, and held back from deployment.
- Rebuilt search ranking as a weighted Postgres full-text function with a title-relevance layer, after measuring that 27.4% of returned results had none of the user's query terms in the job title.
- Hardened the platform: Fernet field encryption of CV text, row-level security across 23 tables, GDPR export and deletion, behind 1,950 automated tests and 11 CI build gates.

Outlyst

Oct 2025 - Mar 2026

AI / Machine Learning Engineer (fixed-term contract)

Leeds, UK (Remote)

- Built and deployed an AI voice-agent system for outbound calling (Retell AI, FastAPI): dialogue-flow logic, automated call flows and lead-qualification rules, handling 2,100+ calls and cutting manual calling workload by roughly 50%.
- Reduced average call latency 54%, 2.4s to 1.1s, by profiling async I/O and restructuring connection pooling.
- Built **VoiceFlow** ([GitHub](https://github.com)), a FastAPI service running Whisper large-v3 in-process over the 2,100+ call recordings, threading each export to keep blocking downloads and torch inference off the asyncio event loop.

COMSATS University Islamabad

Jul 2023 - Sep 2023

Data Science Intern

Islamabad, Pakistan

- Implemented and compared machine learning models for sleep-efficiency prediction in Python (scikit-learn, pandas, NumPy); developed the work into a first-author Springer paper (ICSMAI 2024).

RESEARCH & PROJECTS

FinLaw-UK | [code](#) + [benchmark](#)

2025

MSc Dissertation, University of Bradford

Bradford, UK

- Graph-augmented RAG system for UK financial regulation: hybrid BM25 and BGE-small retrieval fused by RRF, cross-encoder re-ranking, 2-hop Neo4j traversal, and a verifier resolving every citation against the graph. Mistral 7B via Ollama, fully local.
- Found two reported metrics were regex shape-checks, not correctness measures: the scorer awarded a flat 0.85 to any citation-shaped string, and 103 of 110 rows carried that constant. Published the correction and the true graph-verified rate, 3 in 110, which drove the citation normaliser, re-ranker and refusal gate.

DiabetesSense | [code](#)

Jan 2024 - Jun 2024

BSc Final-Year Thesis, COMSATS University Islamabad

Islamabad, Pakistan

- Benchmarked 11 classifiers for diabetes risk on a 253,680-record CDC BRFSS dataset (14% positive class) and deployed the best model as a screening web app (React, Flask) showing a risk percentage and the main contributing factors.

EDUCATION

University of Bradford

Sep 2024 - Sep 2025

MSc, Applied Artificial Intelligence and Data Analytics (Merit)

Bradford, UK

COMSATS University Islamabad

Sep 2020 - Jul 2024

BS, Bioinformatics

Islamabad, Pakistan

PUBLICATIONS

Ahmad, H., Khan, M.U., & Azam, M. (2024). Comparative Analysis of Machine Learning Methods for Enhancing Sleep Efficiency and Prediction. *ICSMAI 2024* (Springer). DOI: [10.1007/978-3-031-66854-8_1](https://doi.org/10.1007/978-3-031-66854-8_1)